

FACULTY OPERATIONS HANDBOOK — 100% COMPREHENSIVE CONTROL & SCREEN REFERENCE

STAFF INSTRUCTOR

COMPLETE OPERATIONS MANUAL

A complete A-to-Z button-by-button, screen-by-screen operations manual for ThoorigAI staff instructors. Covers every interactive element, flow, and modal: OTP onboarding and credential setup, daily authentication, classroom dashboard monitoring, attendance watchlist management, student roster searching and dossier printing, multi-tier student admission, daily live roll-call marking, assessments studio creation, Google Forms & Sheets response synchronization, instant classroom QR code projection, inline evaluation grading studio, cloud learning materials upload and repository management, academic performance analytics, personal account configuration, and classroom emergency procedures.

DOCUMENT SCOPE

Complete Staff Portal Workflow (22 Scenarios • 22 UI Screens)

AUTHORIZED ROLE

Staff Faculty / Instructor (role = staff)

SOFTWARE VERSION

ThoorigAI Academy Core v1.0.0 (Production)

VERIFICATION STATUS

Playwright Audited & DOM-Extracted Live UI

Faculty Governance & Version History

Faculty operational lifecycle controls, version tracking, and academic coordination sign-offs.

Version	Release Date	Author / Role	Summary of Operational Changes
v1.0.0	August 2026	Lead Academic Director	Initial staff manual release covering attendance roll-call and grading studio.
v2.0.0	September 2026	Curriculum & QA Team	DOM-extracted UI flows, classroom QR projection, and Cloud Materials repository.
v2.1.0	October 2026	Faculty Operations Board	Added Faculty Daily Operations Cheat Sheet, Google Forms sync troubleshooting matrix, and mobile/tablet SOPs.
v2.2.0	October 2026	Academic Delivery Team	Added visual classroom pipelines, assessment data dictionary, and Section 29 Classroom Emergency Playbook.

Faculty Daily Operations Cheat Sheet — Quick Reference Card

The 5-step daily operating schedule for instructors and trainers at ThoorigAI Academy.

Time Window	Action Item	Target Module / URL	Operating Instruction & Verification
1 09:00 AM	Morning Roll-Call	/attendance	Select batch → click Mark All Present shortcut → toggle absentees → click Save Attendance (verify green toast).
2 10:00 AM	Publish Daily Assets	/materials	Open assigned course file manager → upload daily lecture slides (PDF) and hands-on lab code (.zip).
3 02:00 PM	Launch Daily Quiz	/assessments	Open Assessment card → click Classroom QR in header → project QR full-screen on lab projector.
4 03:30 PM	Sync Evaluation Marks	/assessments (Studio)	Click Sync Google Sheet or enter inline grades & remarks → click Save All Scores .
5 05:00 PM	Attendance Retention Review	Classroom Dashboard (/)	Review Attendance Watchlist (<75%) → trigger counselor call links for students with 3+ consecutive absences.

Table of Contents

00 — System Architecture & Complete Staff Portal Sitemap	Page 04
01 — Onboarding: Receiving & Validating Single-Use OTP Invite Codes	Page 06
02 — Account Registration & Password Configuration (/signup)	Page 07
03 — Daily Authentication & Multi-Tab Session Security (/login)	Page 08
04 — Global Shell Controls: Topbar, Search & Notifications	Page 09
05 — Staff Classroom Dashboard Overview & Quick Actions Dock	Page 10
06 — Classroom Attendance Health, Watchlist & Faculty SOP Checklist	Page 11
07 — Student Roster Management & Directory Filtering (/students)	Page 12
08 — Student Enrollment & New Admission Modal (/students/new)	Page 13
09 — Comprehensive Student Dossier & Academic History (/students/[id])	Page 14
10 — Daily Attendance Session Picker & Date Controls (/attendance)	Page 15
11 — Live Daily Attendance Marking Sheet (/attendance?courseId=...)	Page 16
12 — Assessments Studio Catalog & Evaluation Overview (/assessments)	Page 17
13 — Creating a New Course Assessment & Google Quiz Setup	Page 18
14 — Google Forms & Sheets Response Import Synchronization	Page 19
15 — Instant Classroom Live QR Code Projection Modal	Page 20
16 — Inline Evaluation & Grading Studio Workflow	Page 21
17 — Course Materials Repository Directory (/materials)	Page 23
18 — Managing Program Materials & Presigned Cloud Downloads	Page 24
19 — Uploading Lesson Materials Modal (/courses/[id]/materials)	Page 25
20 — Academic Reports: Attendance Trends & Date Filters (/reports)	Page 26
21 — Academic Reports: Examination Analytics & Score Rosters	Page 27
22 — Instructor Account & Personal Profile Settings (/settings/user)	Page 28
23 — Custom Profile Metadata & Structured JSON Inspector	Page 29
24 — Complete Staff REST API & Permissions Endpoint Matrix	Page 30
25 — Role Authority Matrix & Faculty Security Boundaries	Page 31
26 — Faculty Keyboard Shortcuts & Rapid Roll-Call Navigation	Page 32
27 — Classroom Hardware, Tablet & Network Matrix	Page 33
28 — Institutional Terminology & Academic Glossary	Page 34
29 — Faculty Classroom Emergency Playbook & Quick-Fix Matrix	Page 35

00 System Architecture & Complete Staff Portal Sitemap

The ThoorigAI Staff Portal is built for faculty, trainers, and academic coordinators. Staff users have complete read/write authority over classroom operations, learner rosters, daily roll-calls, tests, study materials, and academic reporting, while strictly isolating sensitive administrative features.

► Workflow Pipeline: Faculty Classroom Operations & Academic Lifecycle

1. 09:00 Roll-Call
/attendance Batch



2. 10:00 Cloud Assets
Slides & Code



3. 14:00 QR Test
Live Projection



4. 15:30 Grading
Sync & Marks



5. 17:00 Retention
Watchlist (<75%)

1. Classroom Dashboard

/

Academic Command & Action Dock

- Active learners count & batch health
- Today's roll-call progress bar
- At-risk attendance watchlist (<75%)
- Quick action dock for daily routine

2. Students Directory

/students

Learner Records & Admissions

- Search by name, phone, or Register ID
- Filter by course & curriculum category
- Table list and cards grid switcher
- Student Dossier & printable PDF view
- New learner admission registration

3. Attendance Tracker

/attendance

Daily Roll-Call Management

- Batch curriculum selector
- Session calendar & 1-day date shift
- Mark All Present / Absent shortcuts
- Per-student status: Present/Absent/Late/Excused
- Absence remarks & notes capture

4. Assessments Studio

/assessments

Academic Evaluation & Quizzes

- Catalog of tests, assignments & capstones
- Create new assessment form
- Google Forms & Sheets URL linking
- Classroom live QR code projection
- Import Google Form responses via CSV/Sheets
- Inline grading studio with autosave

5. Course Materials

/materials

Cloud Learning Resources Repository

- Directory of courses grouped by tier
- File classification (PDF, Image, Doc, ZIP)
- Presigned cloud storage downloads
- Drag-and-drop file upload modal
- Asset deletion with safety dialog

6. Academic Reports

/reports

Classroom Business Intelligence

- Attendance trends over 30-day window
- Examination average scores & pass rates
- At-risk learner follow-up rosters
- CSV export for external academic audits

7. Account Settings

/settings/user

Instructor Profile Configuration

- Display name & phone contacts
- Department, designation & staff ID
- Faculty bio & teaching background
- Dynamic custom key-value attributes
- JSON profile inspector modal

8. Global Shell Controls

Topbar & Sidebar

Cross-Portal Navigation & Sync

- Sidebar collapse/expand rail toggle
- Global student quick-search input
- Notification bell panel & event feed
- Secure sign out & session termination

i Staff Security Boundaries

Staff members CANNOT access: Billing/Invoices (/invoices), Certificate Issuance (/certificates), Audit Log (/audit-log), User Role Administration (/settings/users), Brand Settings (/settings/brand), or GST Compliance (/settings/gst). Attempting to navigate directly to these routes triggers an automatic HTTP 403 Forbidden redirect.

01 Onboarding: Receiving & Validating Single-Use OTP Invite Codes

To protect institutional security, ThoorigAI accounts cannot be self-registered without authorization. An administrator generates an encrypted, single-use OTP invite code specifically authorized for the Staff role.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Receive Code	Admin generates code via /settings/users → Staff Access	Receive code formatted as STAFF-XXXX-YYYY via official institutional channel	Email / SMS
2	Inspect Prefix	Check that the code prefix is STAFF-	Guarantees account receives faculty permissions upon redemption	Client Verify
3	Open Registration Link	Navigate to https://portal.thoorigai.com/signup?code=STAFF-XXXX-YYYY	System checks code against /api/auth/verify-invite?code=STAFF-XXXX-YYYY	GET /api/auth/verify-invite
4	Validation Check	System confirms code is active, not expired, and not used	Registration form initializes with token locked into form context	200 OK
5	Single-Use Redemption	Complete form and submit registration	Code is permanently consumed upon successful account creation	POST /api/auth/signup

💡 Single-Use Enforcement

Every invite code is cryptographically consumed the instant your registration succeeds. The code cannot be used by any other person or for a second registration.

02

Account Registration & Password Configuration

Route: /signup?code=STAFF-XXXX-YYYY

STAFF ONLY

Complete the staff registration form to initialize your instructor identity. The role is automatically mapped from the invite code token.

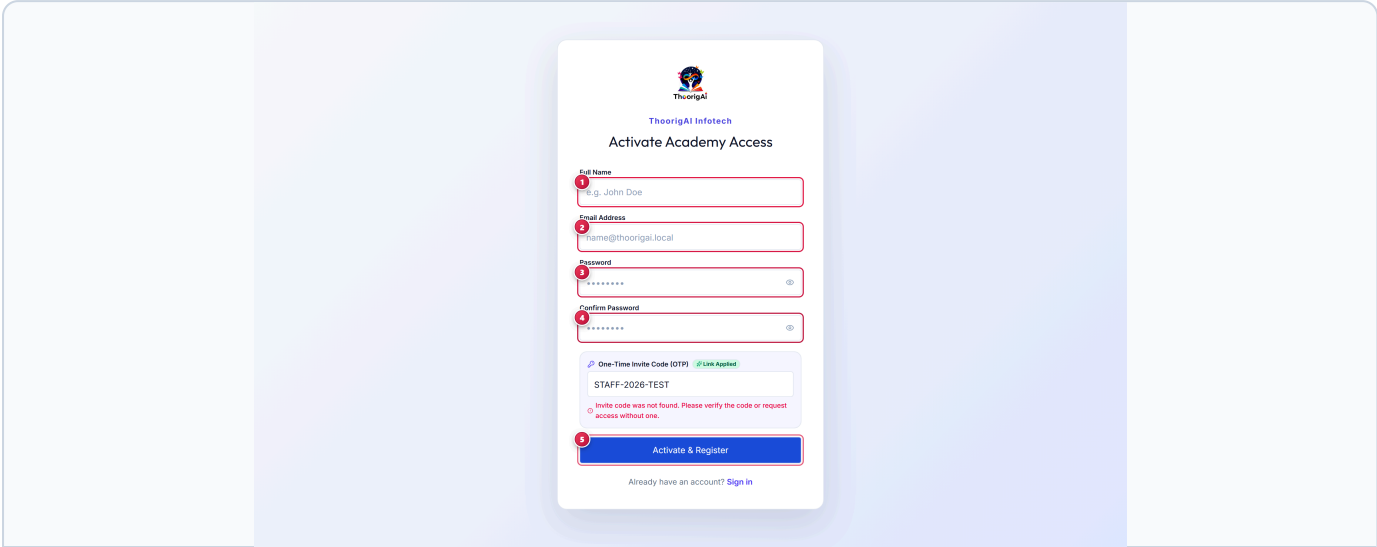


Figure 02 — Account Registration & Password Configuration showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Full Legal Name Field	Enter your full professional name (e.g., "Kavitha Murugesan")	Stored as display name; will appear on course syllabi and grading logs	UI State
2	Institutional Email Field	Enter your official work email address (e.g., "kavitha@thoorigai.edu")	Used as primary authentication credential and for password recovery	UI State
3	Create Secure Password	Type a secure password (minimum 8 characters, letters, numbers, and symbols)	Client verifies password strength; masked for security	UI State
4	Confirm Password Field	Re-enter your password exactly as typed above	Ensures zero typographical errors before database encryption	UI State
5	Complete Registration Button	Click "Complete Registration" to submit	Validates invite code; creates user; sets role=staff; redirects to /	POST /api/auth/signup

03

Daily Authentication Portal & Password Masking

Route: /login

ADMIN + STAFF

Faculty members sign in daily via the authentication portal using their registered email and password.

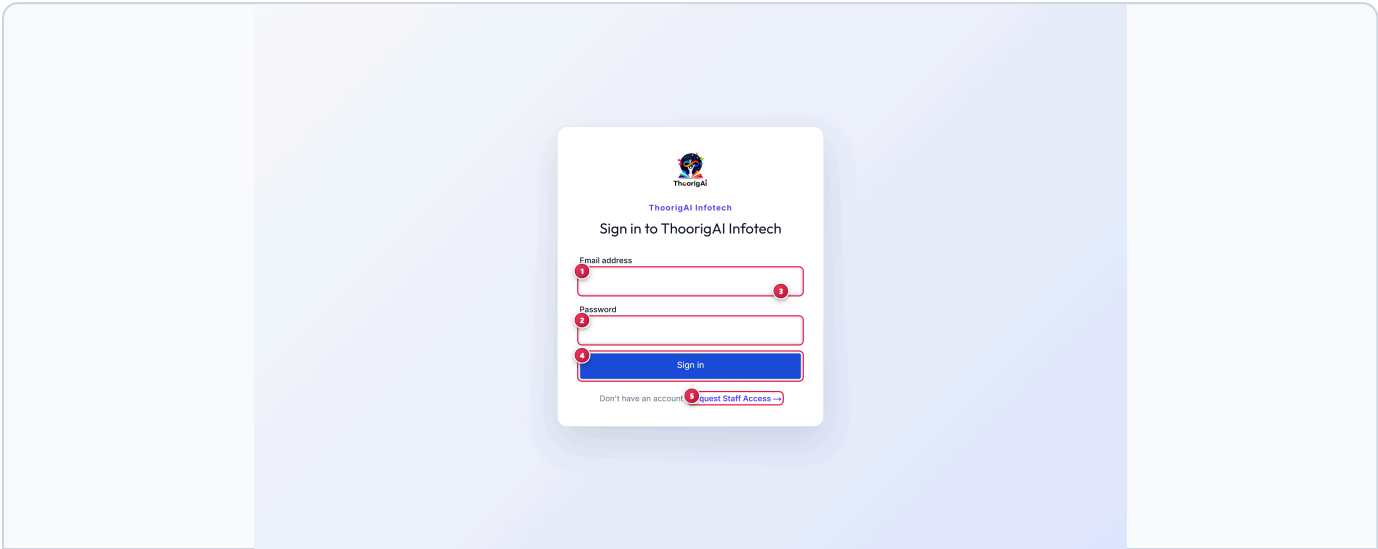


Figure 03 — Daily Authentication Portal & Password Masking showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Faculty Email Address Input	Enter registered faculty email (e.g., staff@thoorigai.test)	RFC 5322 syntax validated in real time; whitespace automatically trimmed	UI State
2	Faculty Account Password Input	Enter confidential account password	Input masked with bullet characters; click eye icon to unmask	UI State
3	Show/Hide Password Toggle	Click eye icon inside password field	Toggles input between masked text and clear text for typo verification	UI State
4	Sign In Submit Button	Click blue "Sign In" button or press Enter	Dispatches POST /api/auth/login; verifies role=staff; routes to /	POST /api/auth/login
5	Request Access / Recovery Link	Click recovery link below form	Directs to self-service recovery; dispatches reset token to registered email	UI State

04

Global Shell Controls: Topbar, Search & Notifications

Route: Global Shell

ADMIN + STAFF

The persistent global shell provides navigation controls, rapid learner lookup, system notification feeds, and secure sign-out controls.

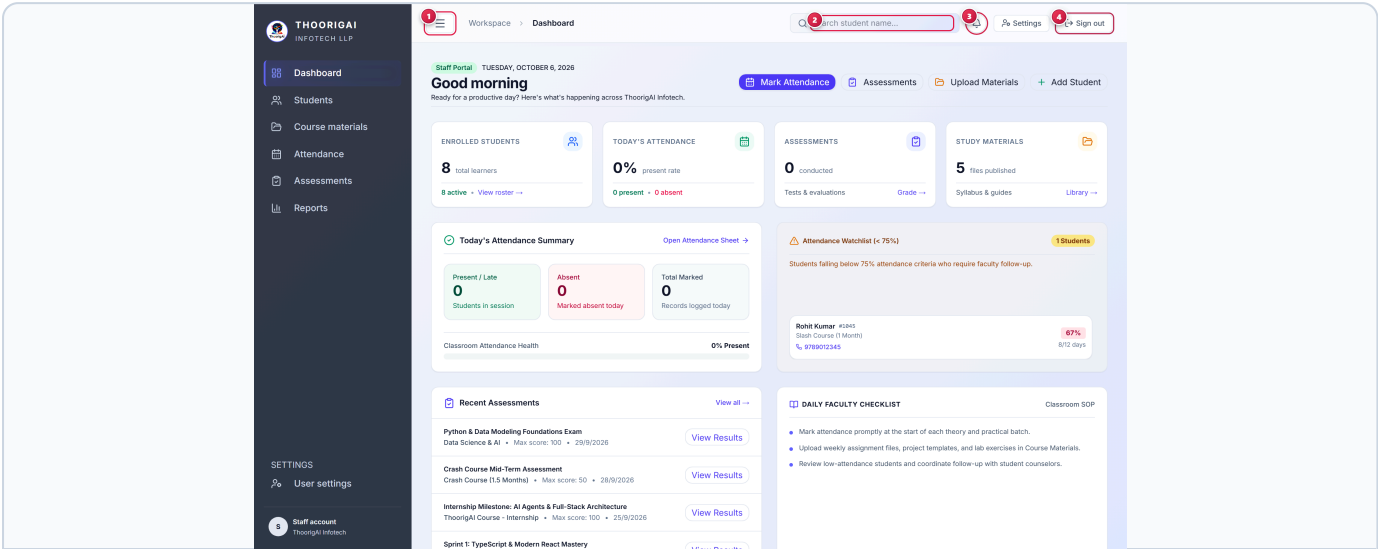


Figure 04 — Global Shell Controls: Topbar, Search & Notifications showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Sidebar Collapse Toggle (Hamburger)	Click triple-line icon at extreme top-left	Collapses sidebar to compact icon rail; click again to expand labels	UI State
2	Global Student Search Input	Click search bar or press "/" hotkey; type name/phone	Queries GET /api/students live; displays matching results in popover	GET /api/students
3	Notification Bell Icon	Click bell icon with unread badge counter	Opens slide-out feed listing batch assignments and attendance alerts	GET /api/notifications
4	Faculty User Avatar & Sign Out	Click "Sign Out" button at top-right	Destroys local session tokens, clears cookies, and routes to /login	POST /api/auth/logout

05

Staff Classroom Dashboard & Quick Actions Dock

Route: /

STAFF ONLY

The central command center for daily academic operations. Provides immediate visibility into active learners, batch completion rates, and roll-call action links.

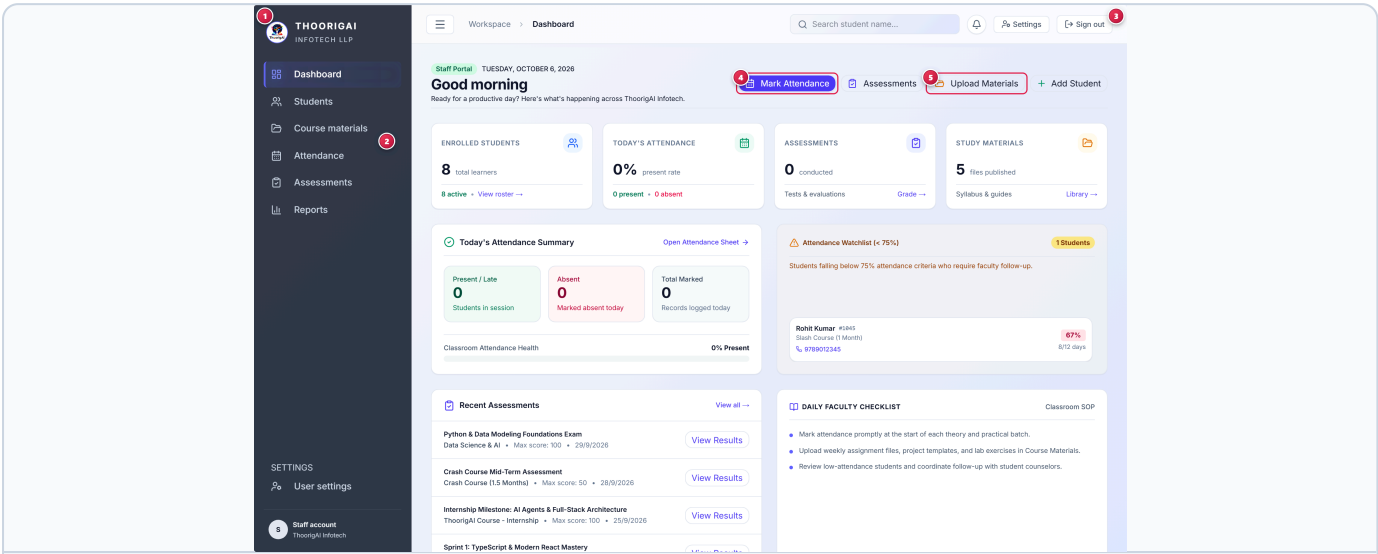


Figure 05 — Staff Classroom Dashboard & Quick Actions Dock showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Welcome Header Banner	Displays greeting with instructor display name and current date	Confirms active logged-in faculty identity and shifts greeting based on time of day	UI State
2	Active Learners Metric Card	Displays total learners enrolled across assigned batches	Summarizes active students under instructor pedagogical supervision	UI State
3	Today's Roll-Call Status Card	Shows percentage and count of students marked for current day	Visual progress bar; turns green (100%) when all assigned batches are submitted	UI State
4	Action Dock: Mark Today's Attendance	Click "Mark Attendance" button with calendar icon	Fast-tracks navigation directly to /attendance with today's date pre-selected	UI State
5	Action Dock: Upload Class Material	Click "Upload File" button with cloud upload icon	Opens the file upload modal directly for currently active course	UI State

06

Classroom Attendance Watchlist & Faculty SOP

Route: / (Dashboard Lower Section)

STAFF ONLY

Automated retention safeguard. Highlights learners falling below the mandatory 75% attendance threshold so instructors can trigger immediate intervention.

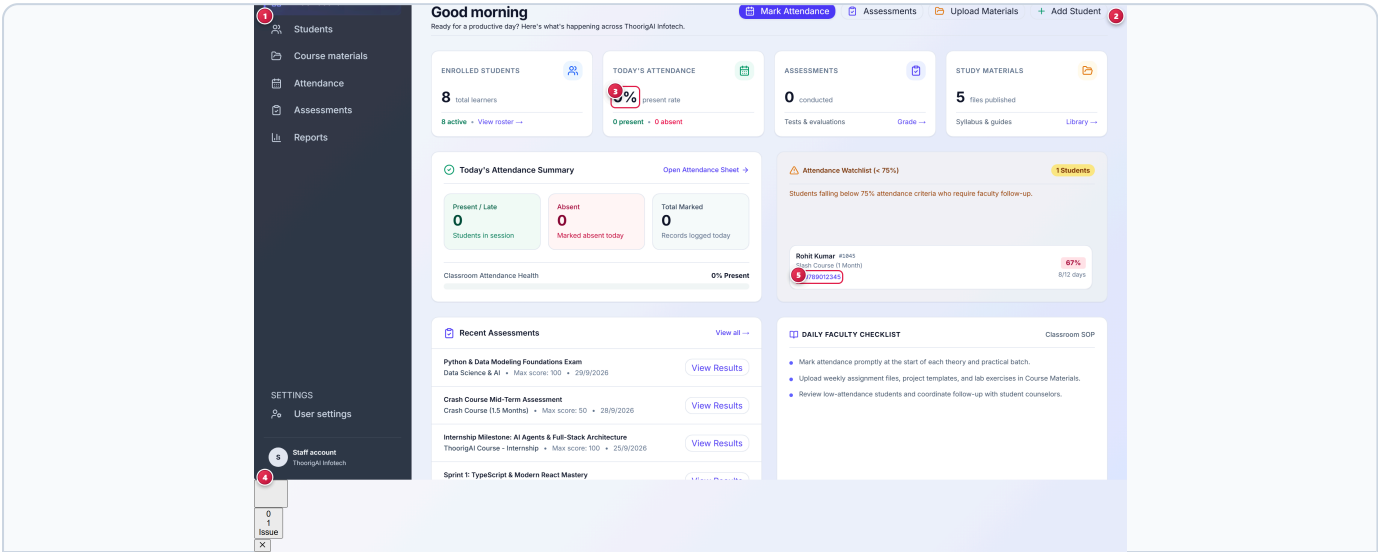


Figure 06 — Classroom Attendance Watchlist & Faculty SOP showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Watchlist Filter Threshold (<75%)	Automatically isolates learners with attendance rate under 75%	Visual red badges highlight at-risk learners needing immediate intervention	UI State
2	Learner Name & Register ID	Shows student identifier (e.g. "Rohit Kumar — #1042")	Click name to open full dossier with historical calendar of absences	UI State
3	Attendance Percentage Badge	Color-coded metric badge (Red: <60%, Amber: 60-74%)	Shows current cumulative attendance percentage for active curriculum	UI State
4	Consecutive Absences Counter	Shows continuous days absent (e.g., "3 days absent")	Alerts faculty if a student has breached the 3-day dropout threshold	UI State
5	Trigger Counselor Call Link	Click phone icon button in row action column	Launches mobile dialer to student guardian; logs intervention note in database	UI State

07

Student Roster Management & Directory Filtering

Route: /students

ADMIN + STAFF

Explore, filter, and inspect learner records. Faculty can view rosters across all active tiers, search by multiple criteria, and switch between table and visual card views.

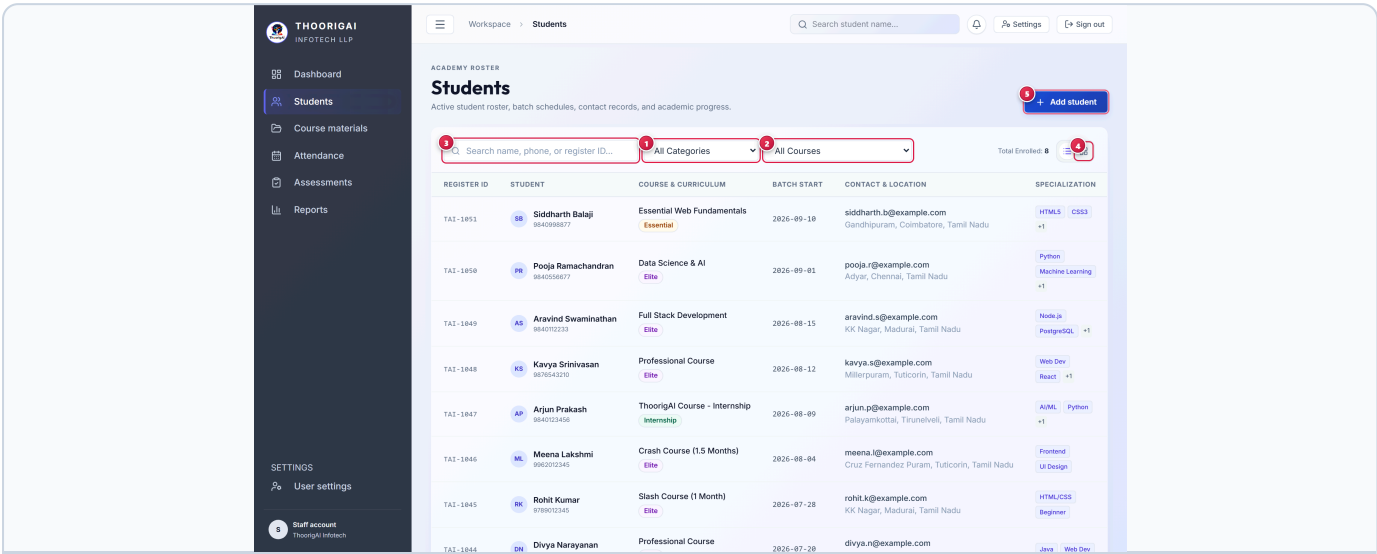


Figure 07 — Student Roster Management & Directory Filtering showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Curriculum Category Selector	Click dropdown: All Categories, Elite, Essential, Internship	Narrows roster to learners enrolled in specific curriculum programs	UI State
2	Course Program Dropdown	Select specific enrolled course (e.g. "Full Stack Development")	Filters roster strictly to students enrolled in that specific syllabus	UI State
3	Multi-Field Search Bar	Type student name, phone number, or Register ID	Real-time client filtering as you type; zero page reload latency	UI State
4	View Switcher: Table / Cards Grid	Toggle between dense data table and visual cards layout	Adapts presentation to instructor preference without resetting filters	UI State
5	New Admission Button	Click "+ Add Student" button at top-right	Opens the New Student Admission registration form	UI State

08

Student Enrollment & New Admission Form

Route: /students/new

ADMIN + STAFF

Faculty members authorized by administration can enroll incoming learners with academic course tier and contact information.

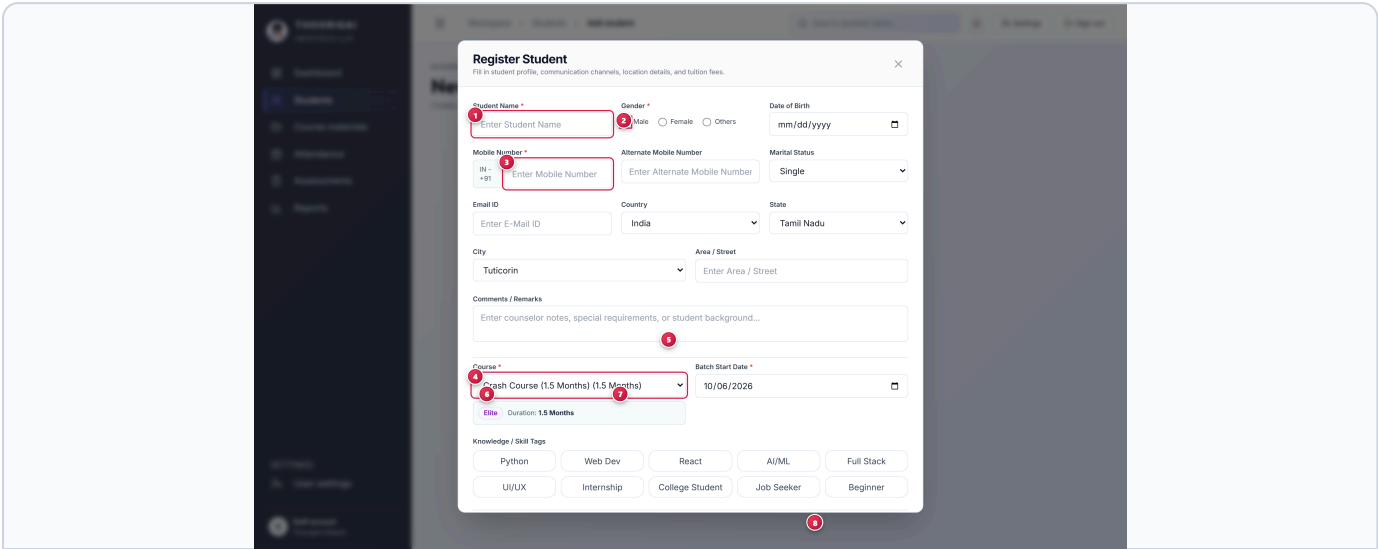


Figure 08 — Student Enrollment & New Admission Form showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Student Legal Full Name Input*	Enter student's official name as printed on certificates	Required; minimum 2 characters; sanitized against special characters	UI State
2	Gender Radio Selector*	Select Male, Female, or Other	Required demographic categorization for state educational reporting	UI State
3	Primary Mobile Phone Input*	Enter 10-digit mobile number (e.g., 9840123456)	Validated against Indian mobile number regex (starts with 6, 7, 8, or 9)	UI State
4	Course Curriculum Dropdown*	Select target course from active catalog	Associates student with correct syllabus and batch attendance sheet	UI State
5	Batch Calendar Start Date*	Select cohort start date from calendar picker	Establishes enrollment anniversary for certificate eligibility	UI State
6	Total Tuition Fee Input*	Enter agreed total course fee in INR	Initializes master invoice record in Supabase billing table	UI State
7	Initial Amount Paid Input	Enter initial deposit paid during admission	If greater than 0, generates payment receipt; remainder tagged as pending due	UI State
8	Submit Registration Button	Click "Register Student" at bottom of modal	Calls POST /api/students; creates student & invoice; displays success toast	POST /api/students

09

Comprehensive Student Dossier & Academic History

Route: /students/[id]

ADMIN + STAFF

Complete 360-degree academic record for an individual learner. Displays personal demographics, attendance history calendar, evaluation results, and printable transcript view.

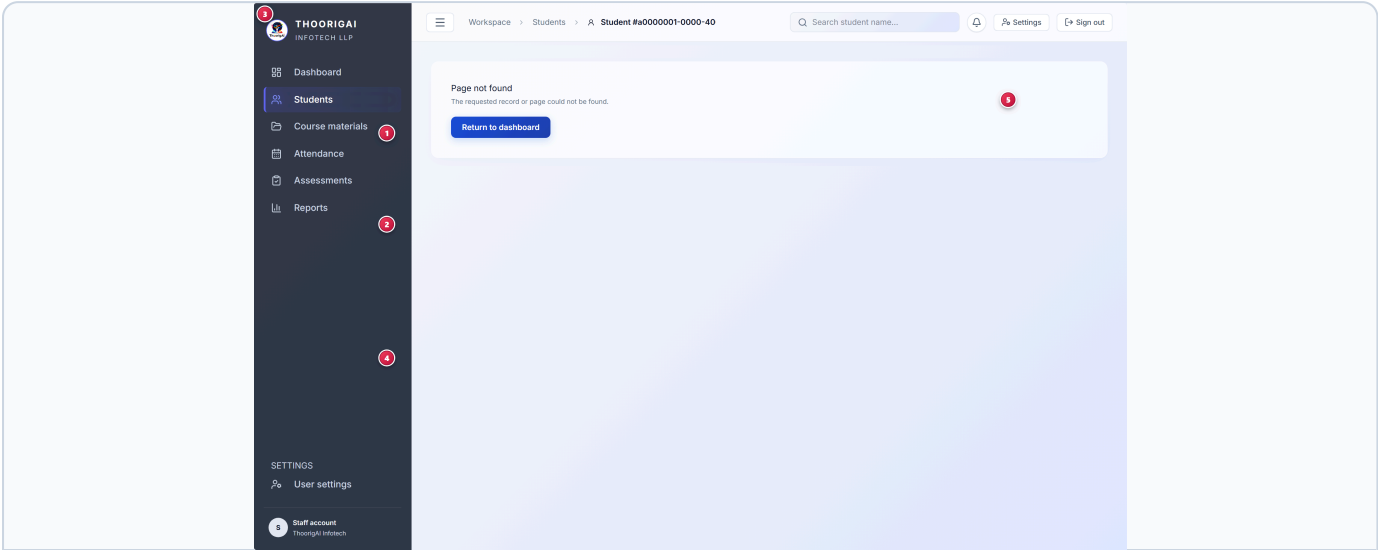


Figure 09 — Comprehensive Student Dossier & Academic History showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Learner Profile Banner	Displays student photo/avatar, legal name, Register ID, and status badge	Visual summary of student identity and active/completed academic standing	UI State
2	Contact & Personal Info Card	Shows phone number, email address, enrollment date, and enrolled tier	Quick reference for student communication and batch coordinator oversight	UI State
3	Attendance Performance Card	Shows total sessions, days present, days absent, and percentage score	Provides instant diagnosis of attendance compliance against 75% threshold	UI State
4	Weekly Attendance Breakdown	Displays daily breakdown across recent training weeks	Reveals weekly consistency and identifies unexcused absence patterns	UI State
5	Print Dossier PDF Action	Click "Print Dossier" button at top-right of page	Invokes browser print layout with clean formatting suitable for academic records	UI State

10

Daily Attendance Session Picker & Date Controls

Route: /attendance

ADMIN + STAFF

Configure and target the specific batch and date session for attendance marking. Features one-click date navigation and historical audit records.

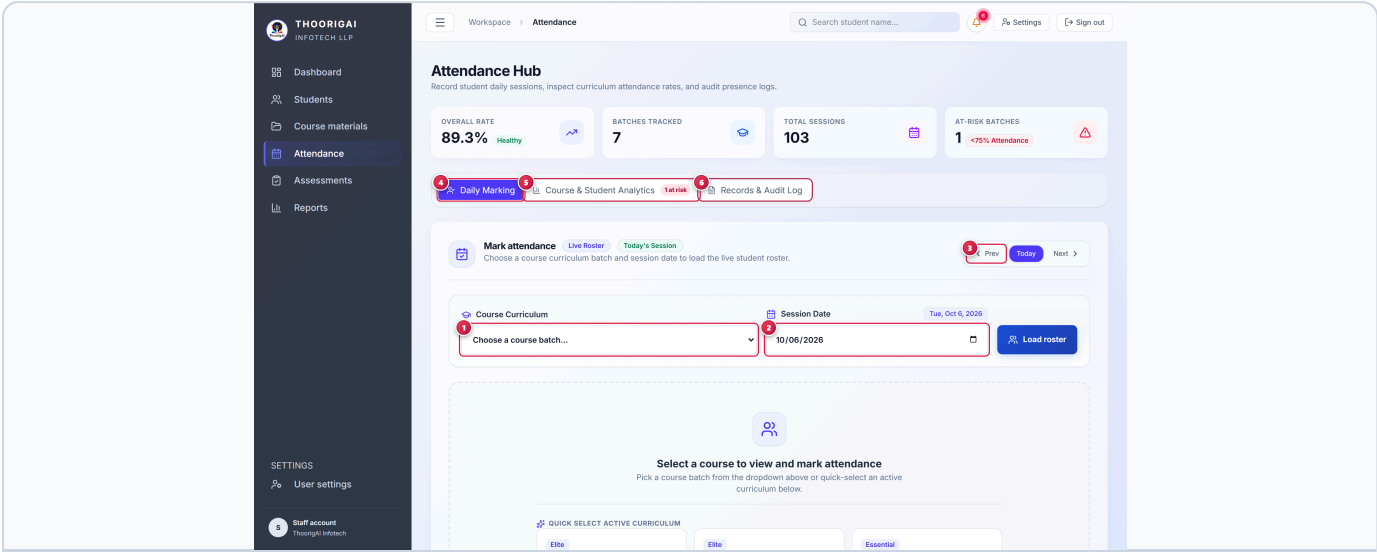


Figure 10 — Daily Attendance Session Picker & Date Controls showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Curriculum Batch Dropdown	Select course from grouped dropdown (Elite, Essential, Internship)	Filters the live student roster to only learners enrolled in that specific batch	UI State
2	Session Date Picker	Select the date for roll-call (defaults to today's date)	Loads previous marks if already taken, or fresh blank roster if taking first time	UI State
3	Quick Date Shift Controls	Click Prev, Today, or Next buttons	Instantly shifts date by one day without opening date calendar picker	UI State
4	Daily Marking Tab	Default tab for active roll-call input	Displays the batch selection form and live marking sheet	UI State
5	Course & Student Analytics Tab	Click "Course & Student Analytics" tab	Switches to aggregate batch-level attendance percentages and at-risk metrics	UI State
6	Records & Audit Log Tab	Click "Records & Audit Log" tab	Displays historical roll-call logs with timestamp and instructor attribution	UI State

11 Live Daily Attendance Marking Sheet

Route: /attendance?courseId=...&date=...

ADMIN + STAFF

The live roll-call worksheet with active batch roster loaded. Features one-click bulk marking shortcuts, individual student status toggles, and qualitative absence remarks.

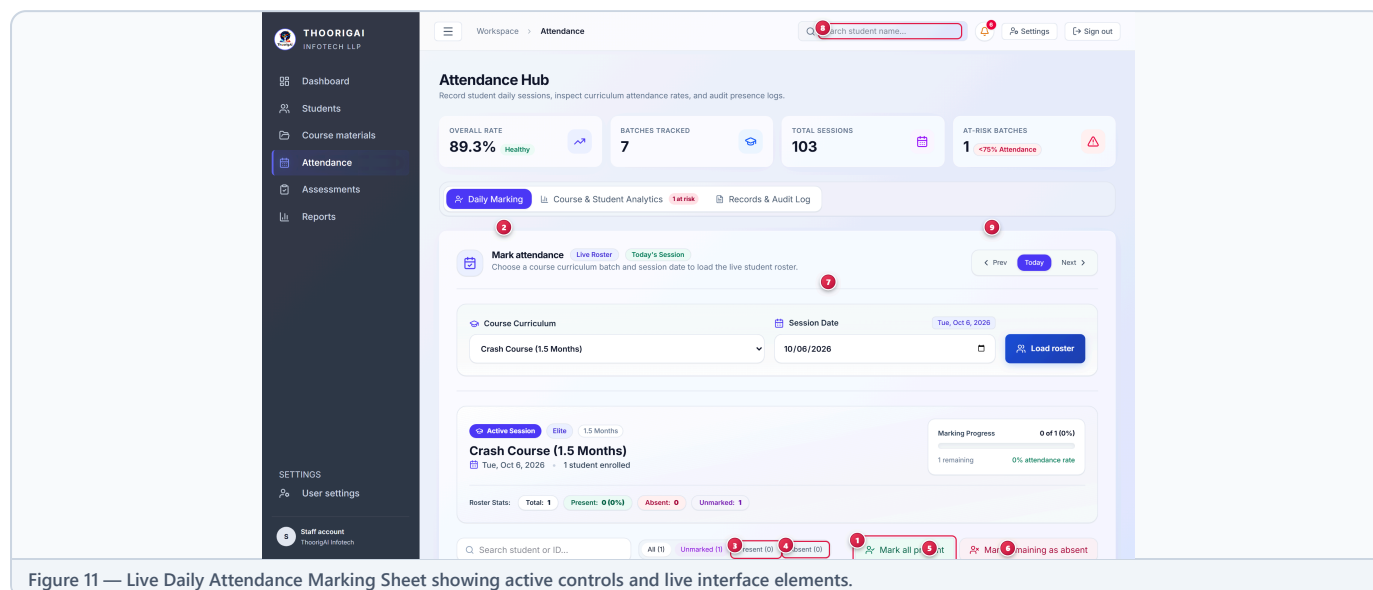


Figure 11 — Live Daily Attendance Marking Sheet showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Bulk Action: Mark All Present	Click "Mark All Present" button (green checkmark icon)	Instantly sets every student in the batch roster to "Present" in a single click	UI State
2	Bulk Action: Mark All Absent	Click "Mark All Absent" button (red cross icon)	Sets every student to "Absent"; ideal for unexpected batch holidays	UI State
3	Status Button: Present (Green)	Click "Present" in a student row	Marks student present for both theory and lab hours	UI State
4	Status Button: Absent (Red)	Click "Absent" in a student row	Records unexcused absence; updates student absence total	UI State
5	Status Button: Late (Amber)	Click "Late" in a student row	Records student arrived after grace period; counted as present with tardiness flag	UI State
6	Status Button: Excused (Blue)	Click "Excused" in a student row	Records sanctioned absence (medical leave, approved campus event)	UI State
7	Faculty Remarks / Reason Input	Type notes in the "Remarks / Note" text field per student	Document reasons for absence or late arrival; persisted with attendance record	UI State
8	Filter Roster Search Box	Type in the inline search field	Instantly filters the displayed roster by student name or Register ID	UI State
9	Save Attendance Commit Button	Click blue "Save Attendance" button at bottom of sheet	Calls POST /api/attendance; executes bulk upsert in Supabase; shows success toast	POST /api/attendance

12

Assessments Studio Catalog & Evaluation Overview

Route: /assessments

ADMIN + STAFF

Manage and conduct examinations, quizzes, and project evaluations. Displays high-level examination KPIs, search filters, and action cards for every assessment.

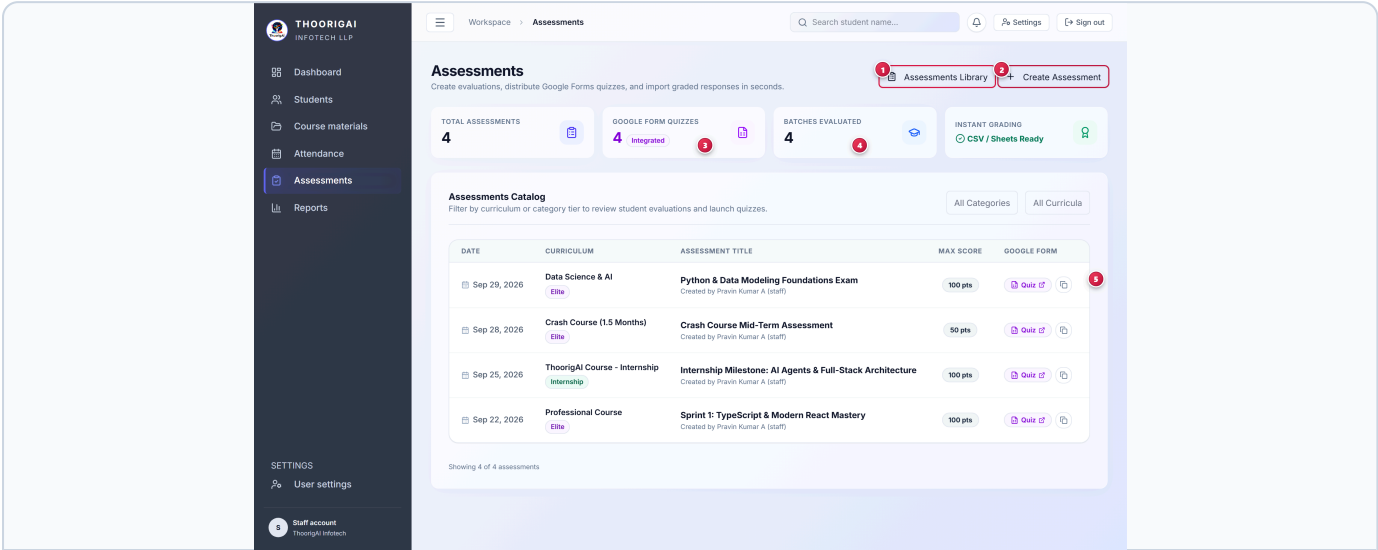


Figure 12 — Assessments Studio Catalog & Evaluation Overview showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Assessments Library Tab	Default view listing all created evaluations	Shows active tests with title, course badge, maximum score, and conduct date	UI State
2	Create Assessment Tab	Click "+ Create Assessment" button at top right	Switches view to the assessment creation form	UI State
3	Search Filter Input	Type assessment title keywords in search box	Filters catalog cards to matching test titles	UI State
4	Curriculum Category Filter	Select tier (All Categories / Specific Tier)	Limits displayed assessments to programs within that category	UI State
5	Open Studio Action Button	Click "Open Studio" on any assessment card	Launches the dedicated Grading & Results Studio for that evaluation	UI State

13 Creating a New Course Assessment & Google Quiz Setup

Route: /assessments (Create Tab)

ADMIN + STAFF

Create an evaluation for an assigned course. Link Google Forms quizzes or Google Sheets to automatically generate live student test links and QR projection codes.

Figure 13 — Creating a New Course Assessment & Google Quiz Setup showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Assessment Title Input	Enter descriptive name (e.g. "Week 3: React Hooks & State")	Title displayed on student reports, grade sheets, and notifications	UI State
2	Target Curriculum Combobox	Click dropdown and select enrolled course program	Links test to the course; populates enrolled learners into grading studio	UI State
3	Max Score (Total Points)	Enter maximum possible points (e.g., 30, 50, 100)	Used to calculate percentage and determine pass/fail threshold	UI State
4	Assessment Date Picker	Select scheduled date of examination	Establishes official test date on student dossiers and academic transcripts	UI State
5	Google Form Student URL	Paste student Google Form quiz link	Enables one-click Classroom QR code projection and direct form launch	UI State
6	Google Sheets Response URL	Paste Google Sheets spreadsheet link where form responses collect	Enables direct opening of responses and automated grade importing	UI State
7	Create Assessment Submit Button	Click "Create Assessment" button	Calls POST /api/assessments; creates assessment; redirects to library	POST /api/assessments

14

Google Forms & Sheets Response Import Synchronization

Route: Modal Component

ADMIN + STAFF

Synchronize test scores directly from Google Sheets or Google Forms response spreadsheets with automated column matching.

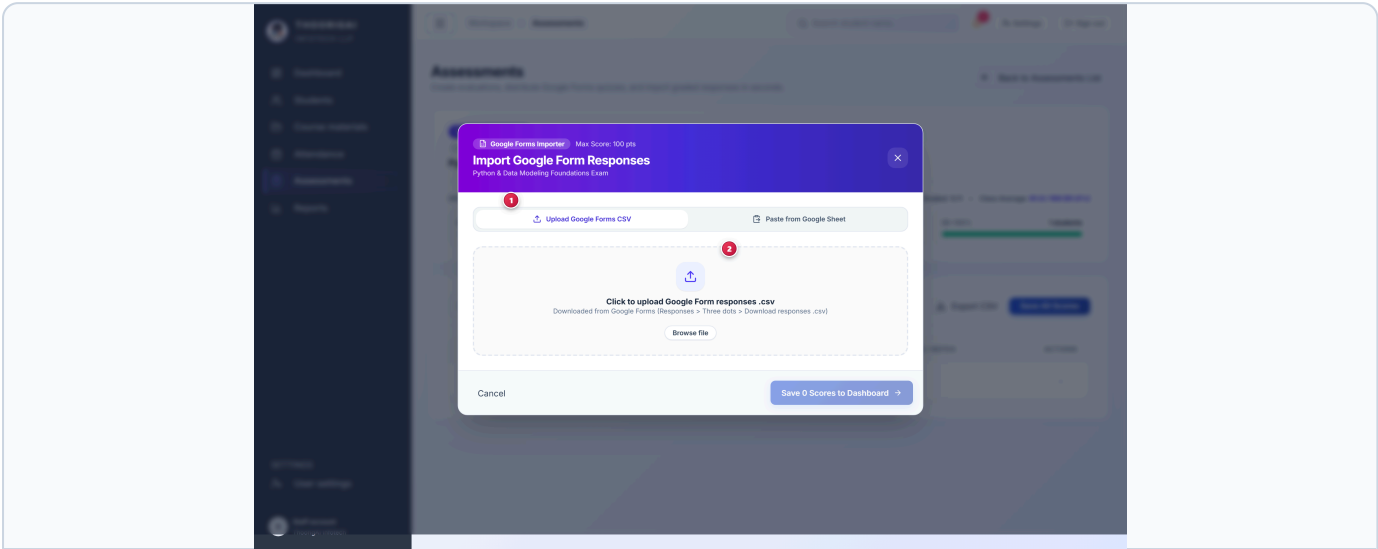


Figure 14 — Google Forms & Sheets Response Import Synchronization showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Google Sheet Published URL Input	Paste Google Sheets spreadsheet URL containing test responses	Validates URL syntax; checks for public or shared sheet access	UI State
2	Sync Responses Now Action Button	Click "Sync Responses Now" button	Fetches spreadsheet rows; parses score columns; populates table	UI State

15

Instant Classroom Live QR Code Projection Modal

Route: Modal Component

ADMIN + STAFF

Instant full-screen projection of high-contrast QR codes directly onto classroom projectors for zero-friction student quiz access.

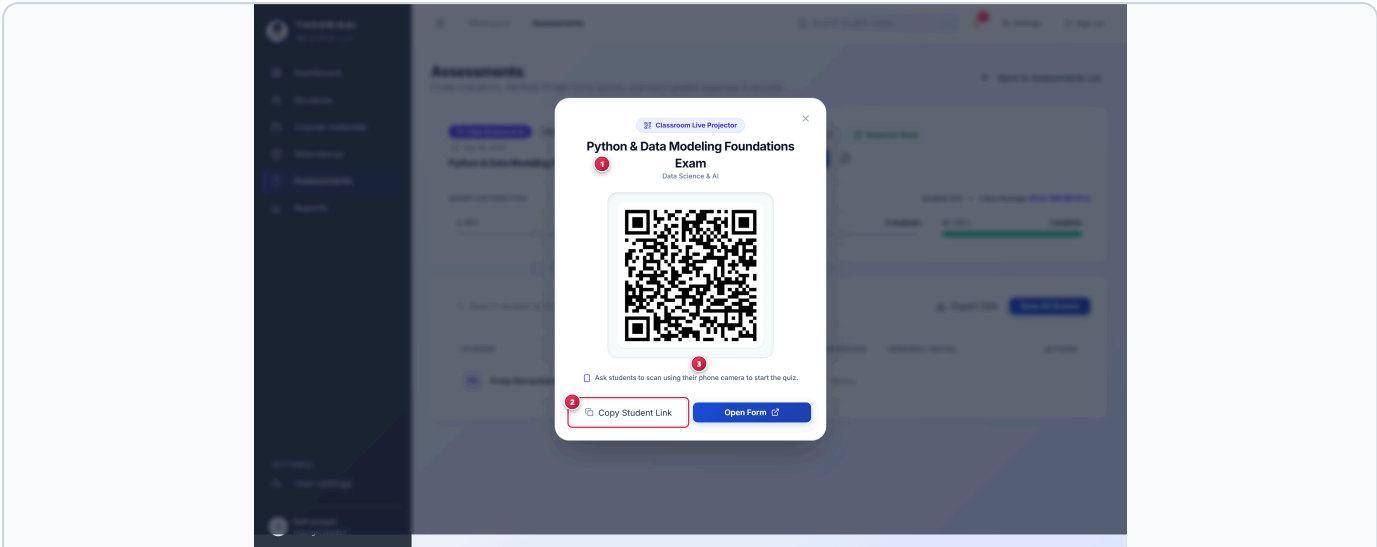


Figure 15 — Instant Classroom Live QR Code Projection Modal showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	High-Resolution Vector QR Matrix	Generated automatically from the student Google Form quiz link	Scannable by all student smartphone cameras at up to 10 meters distance	UI State
2	Copy Direct Quiz Link Button	Click "Copy Link" button	Copies URL to system clipboard; shows confirmation toast	UI State
3	Project Fullscreen Button	Click "Fullscreen" button	Expands QR code to fill entire display for auditorium projection	UI State

16

Inline Evaluation & Grading Studio Workflow

Route: /assessments (Grading Studio)

ADMIN + STAFF

Score student evaluations directly in the browser with inline grading cells, qualitative remarks, and real-time pass/fail metric calculations.

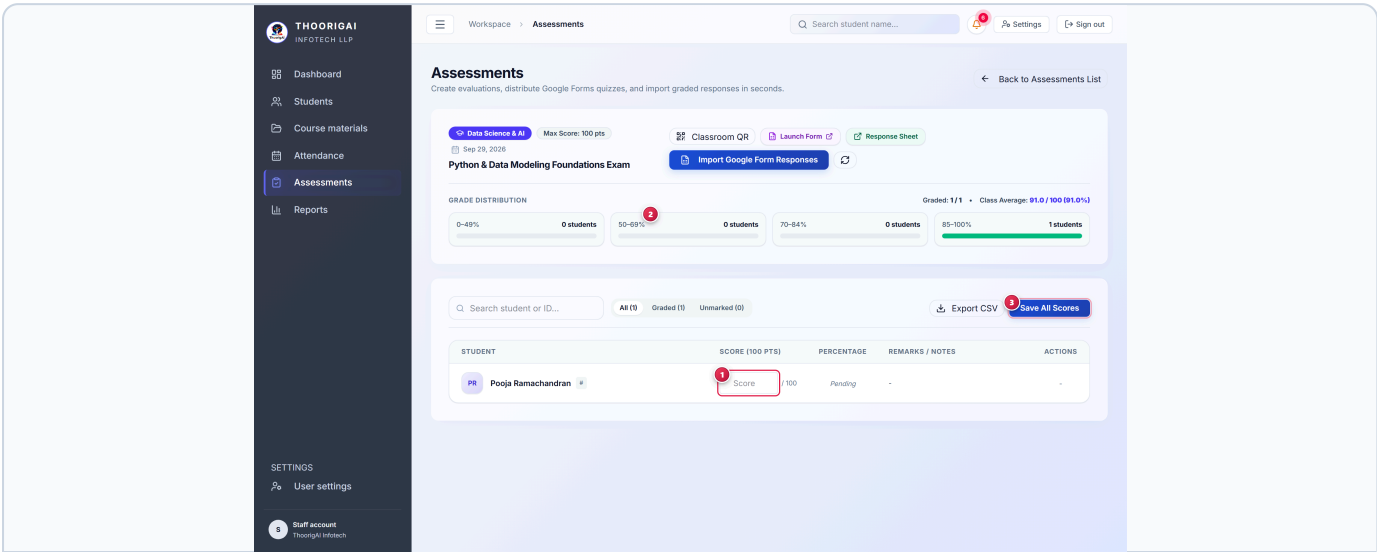


Figure 16 — Inline Evaluation & Grading Studio Workflow showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Inline Score Input Field	Type student points earned (e.g. 94)	Validates score does not exceed Max Score; recalculates % live	UI State
2	Qualitative Remarks Input	Enter constructive feedback for the learner	Saved to student evaluation transcript and visible in dossier	UI State
3	Save All Scores Commit Button	Click "Save All Scores" button	Dispatches bulk upsert to assessment_results table; triggers sync toast	POST /api/assessments/[id]/grades



Field Name	Data Type	Validation Rule	Operational Usage
title	String (1-100 chars)	Mandatory; non-empty	Assessment name displayed on student cards & reports.
max_score	Integer (1-500)	Positive integer	Denominates total marks available for pass/fail calculation.
passing_score	Integer ($\leq$ max)	Must not exceed max_score	Threshold for passing status and certificate eligibility.
google_form_url	Valid HTTPS URL	Must point to docs.google.com	Source quiz embedded in student QR code generator.
responses_sheet_url	Valid HTTPS URL	Shared Google Sheets link	Source spreadsheet parsed during automated response import.

17

Course Materials Repository Directory

Route: /materials

ADMIN + STAFF

Browse courses to access learning resources, lecture slides, code repositories, and lab documentation.

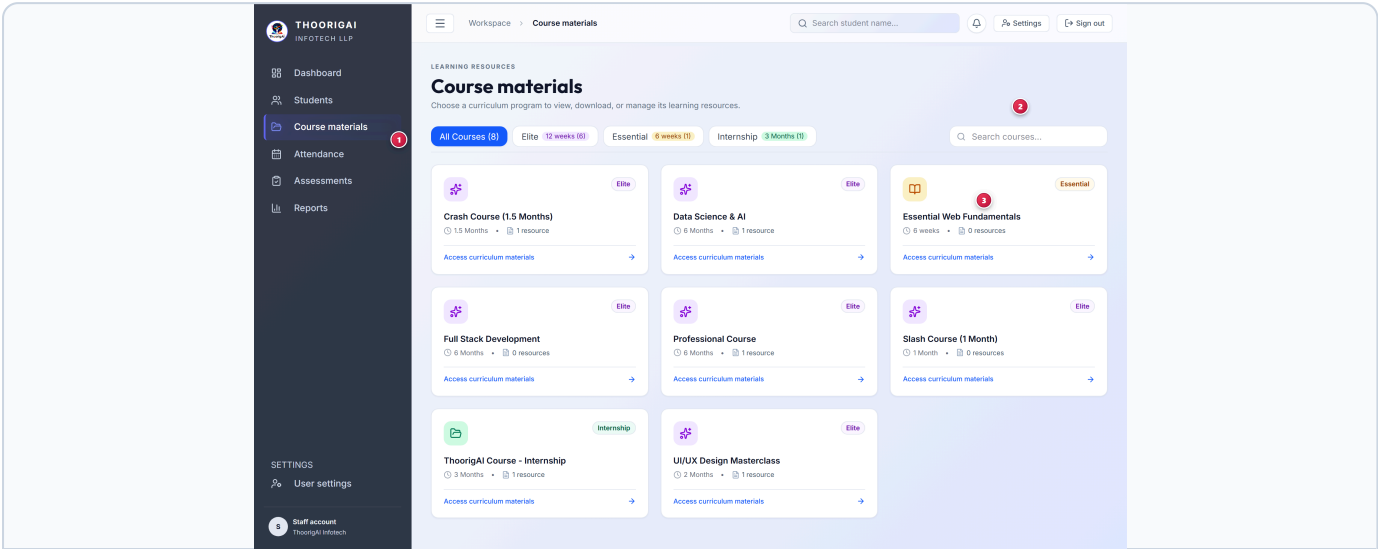


Figure 17 — Course Materials Repository Directory showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Materials Search Input	Type lesson title, file format, or course name	Filters learning resources directory in real time	UI State
2	Upload New Material Button	Click "Upload File" button in header	Opens the cloud document upload modal with drag-and-drop file target	UI State
3	Open Course Folder Button	Click "Open" button on any course card	Navigates to dedicated file manager for that specific course	UI State

18

Managing Program Materials & Presigned Cloud Downloads

Route: /courses/[id]/materials

ADMIN + STAFF

Inspect, download, and manage study materials uploaded to an assigned curriculum program with presigned cloud storage URLs.

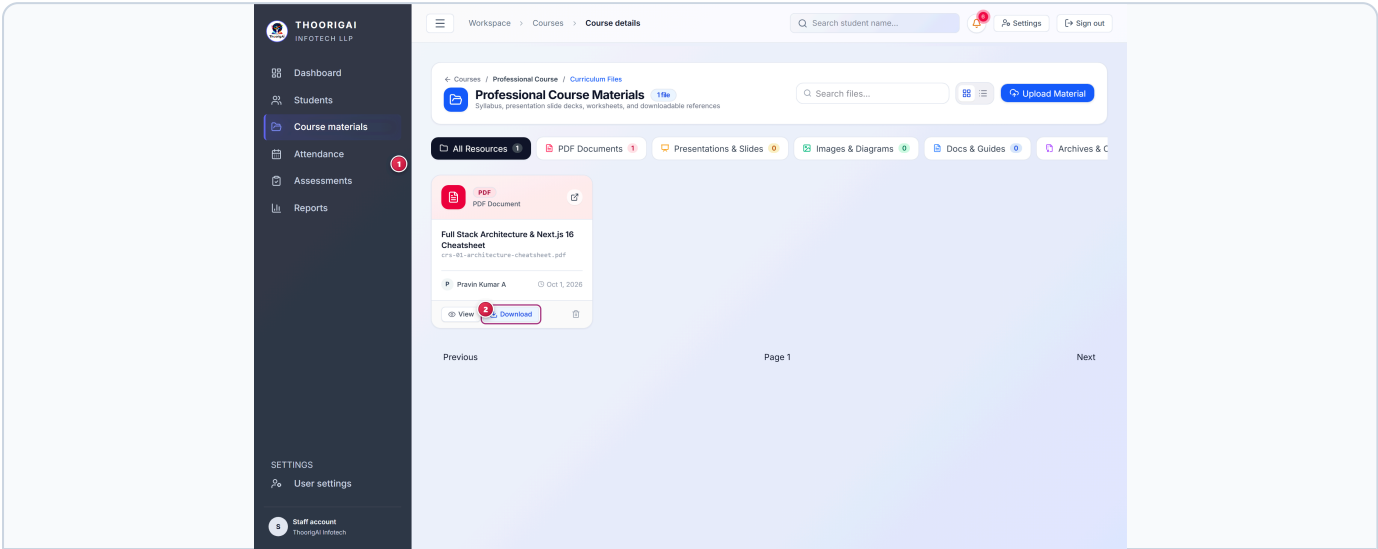


Figure 18 — Managing Program Materials & Presigned Cloud Downloads showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Course Files & Assets Table	Inspect files uploaded to selected course	Displays file title, MIME type badge, file size, and upload date	UI State
2	Download Presigned Cloud Asset	Click download button on any material card	Fetches presigned download URL from cloud storage bucket and saves file	UI State

19

Uploading Lesson Materials Modal

Route: Modal Component

ADMIN + STAFF

Upload slides, lab guides, and reference documents (up to 50 MB per file) with instant cloud streaming and MIME type verification.

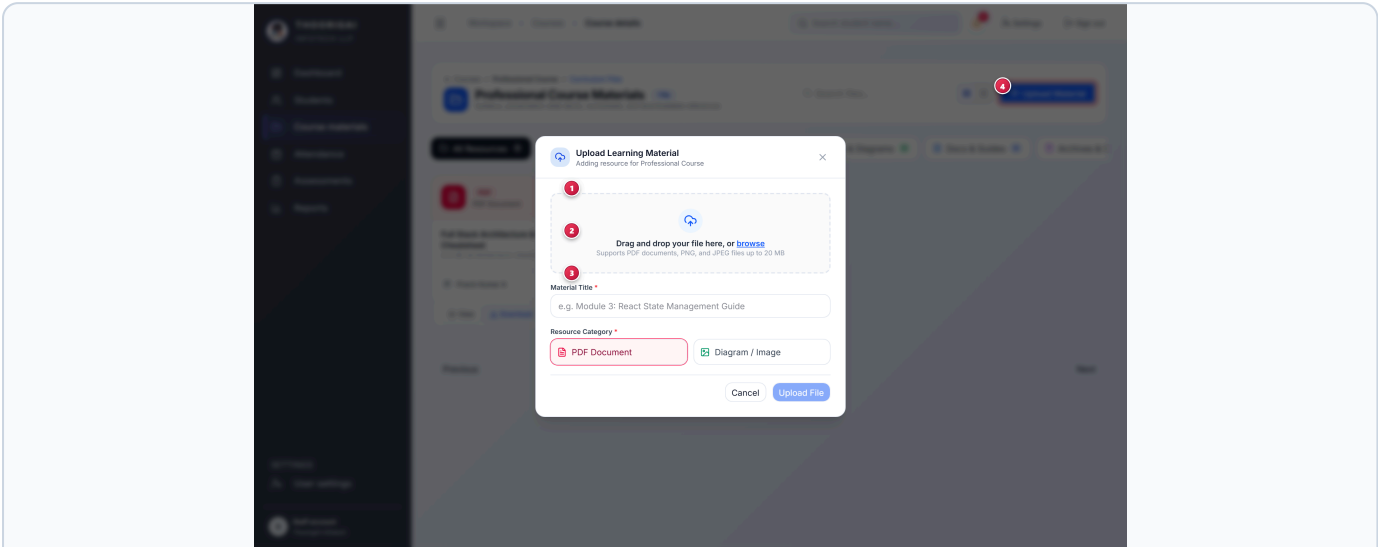


Figure 19 — Uploading Lesson Materials Modal showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Material Title Input	Type descriptive title for resource (e.g., "Lecture 4 Slides - React State")	Displayed in student and faculty materials catalogs	UI State
2	Target Course Selector	Select course program to receive file	Associates asset with syllabus; grants download access to enrolled learners	UI State
3	Drag & Drop Cloud File Target	Drag file into target area or click to select from local storage	Validates file type (PDF, DOCX, PPTX, ZIP, MP4) and verifies size is under 50 MB limit	UI State
4	Commit Upload Submit Button	Click "Upload Material" button	Streams file to Supabase storage; creates metadata row; shows success toast	POST /api/materials

20

Academic Reports: Attendance Trends & Date Filters

Route: /reports

ADMIN + STAFF

Analyze batch attendance compliance over customizable date ranges with interactive trend graphs and CSV exports.

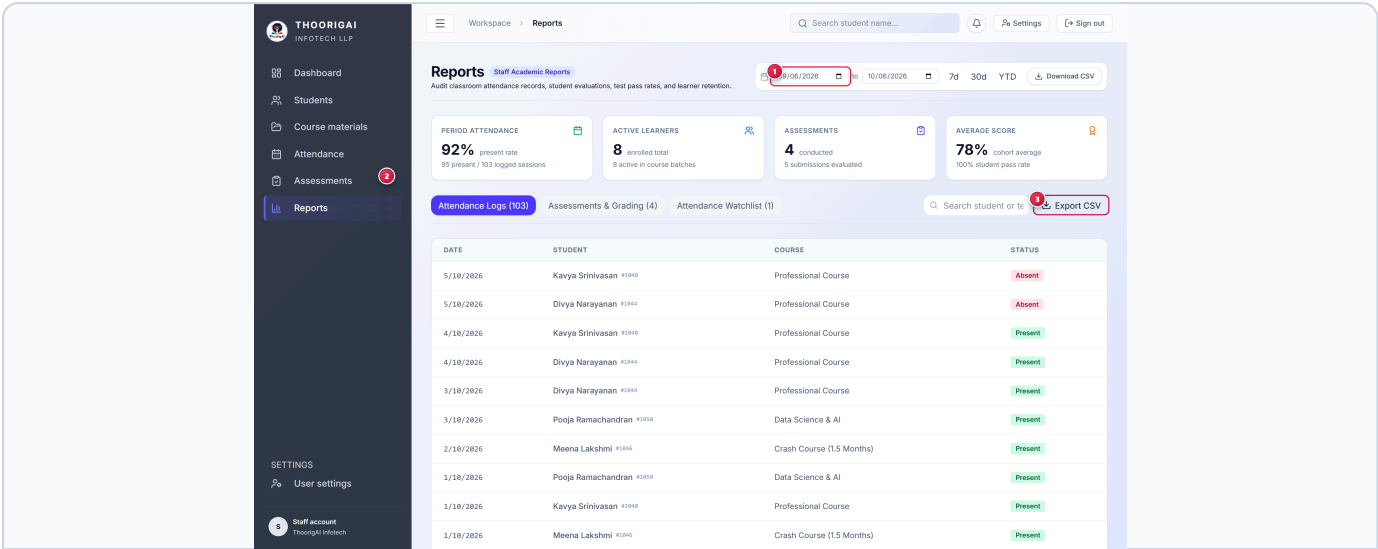


Figure 20 — Academic Reports: Attendance Trends & Date Filters showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Date Range Window Selector	Pick start and end dates to focus analysis	Recalculates attendance and examination metrics for specified time window	UI State
2	Cohort Attendance Rate Graph	Analyze daily roll-call adherence trends across batches	Visual representation of student presence trends across cohorts	UI State
3	Export Academic Report CSV	Click "Export CSV" button	Downloads comprehensive CSV spreadsheet with per-student attendance rates	UI State

21

Academic Reports: Examination Analytics & Score Rosters

Route: /reports (Assessments Tab)

ADMIN + STAFF

Review examination score averages, pass rates, and grade distribution across all completed tests and quizzes.

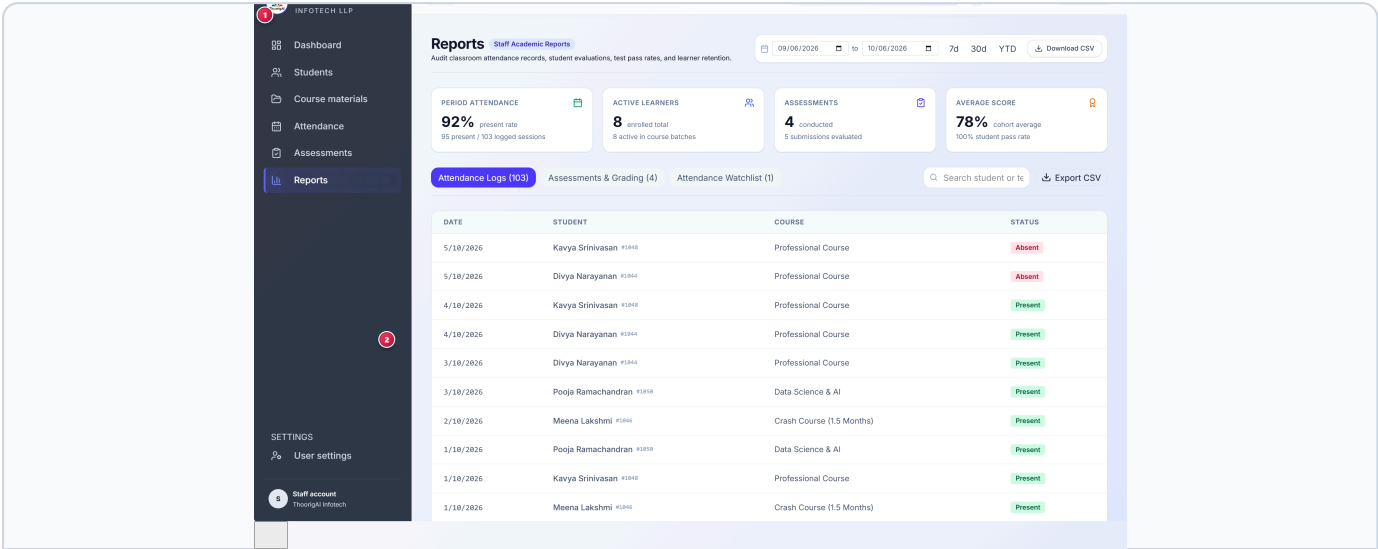


Figure 21 — Academic Reports: Examination Analytics & Score Rosters showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Institutional Examination Average Card	Inspect mean score across all evaluations	Benchmark metric for curriculum effectiveness and batch learning outcomes	UI State
2	Examination Performance Table	Review test-by-test breakdown with candidate counts and pass rates	Displays assessment titles, conduct dates, enrolled cohorts, and top performers	UI State

22

Instructor Account & Personal Profile Settings

Route: /settings/user

ADMIN + STAFF

Update faculty display name, contact phone number, and teaching bio. Display name is stamped on grading transcripts.

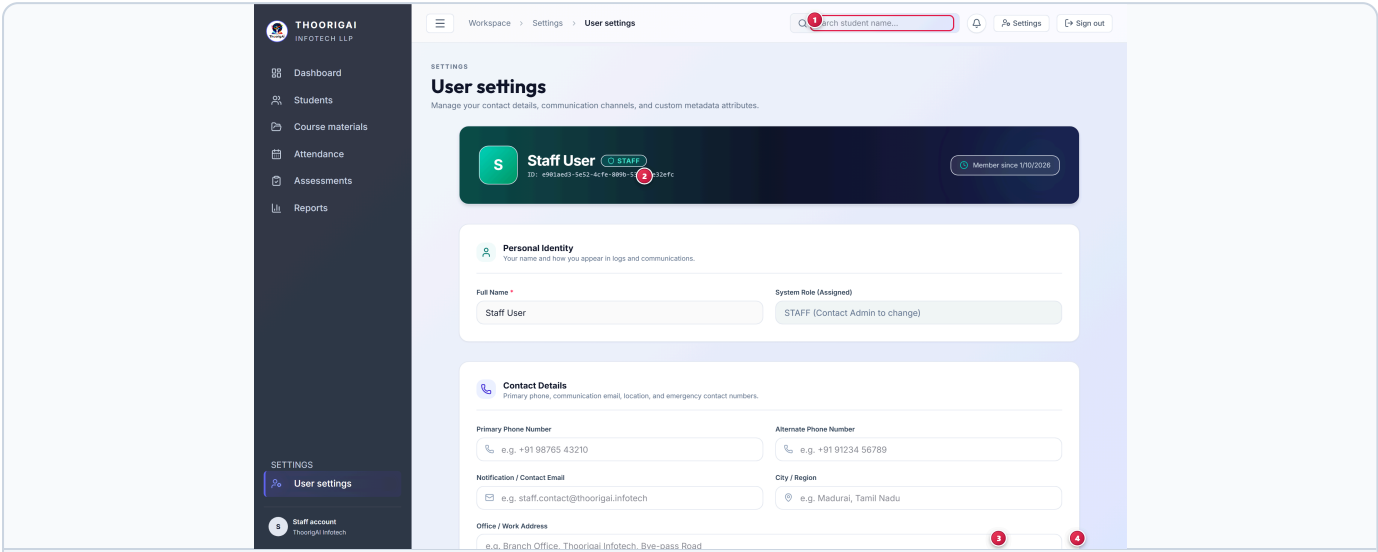


Figure 22 — Instructor Account & Personal Profile Settings showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Faculty Display Name Field	Enter preferred display name (e.g. "Pravin Kumar A")	Used on course syllabi, attendance sheets, and grading signatures	UI State
2	Contact Phone Number Field	Enter emergency contact mobile number	Stored in profiles table for institutional communication	UI State
3	Add Custom Attribute Button	Click "+ Add Field" button	Adds dynamic key-value attribute pair to profile JSON metadata	UI State
4	Save Profile Changes Button	Click "Save Changes" button	Calls PATCH /api/users/[id]; updates profile; shows confirmation toast	PATCH /api/users/[id]

23

Custom Profile Metadata & Structured JSON Inspector

Route: /settings/user

ADMIN + STAFF

Manage dynamic custom key-value profile attributes and inspect the live structured JSON representation.

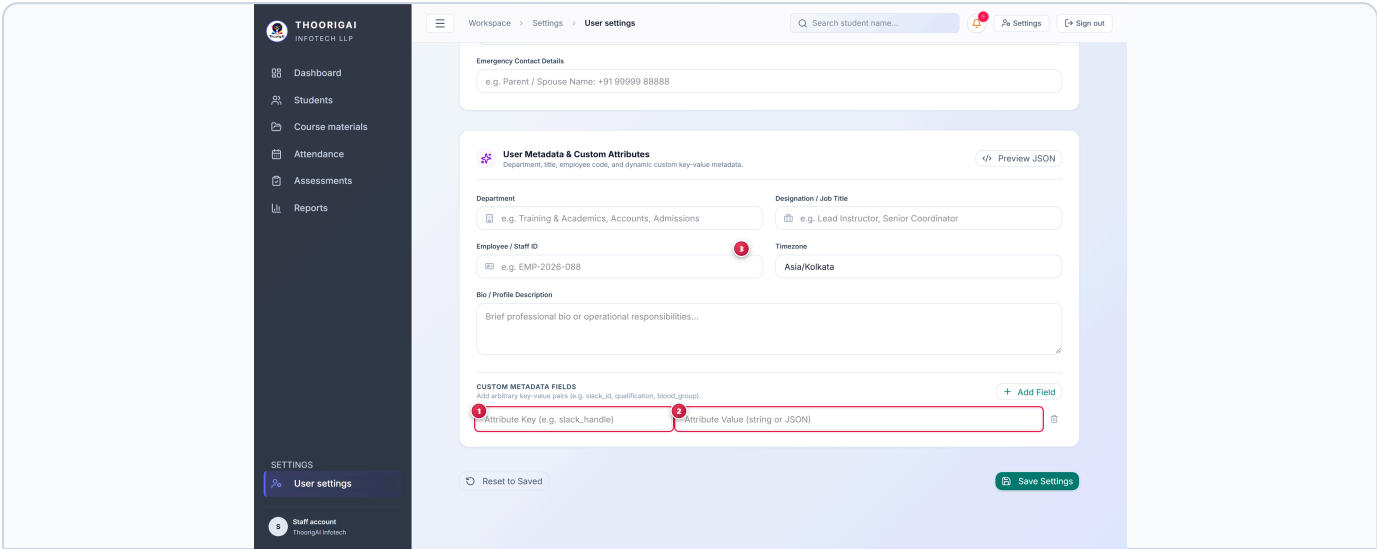


Figure 23 — Custom Profile Metadata & Structured JSON Inspector showing active controls and live interface elements.

Step	Control / Action	How to Use / Details	System Outcome	API / Route
1	Attribute Key Field	Enter attribute key (e.g. "Department", "Office Hours", "Specialization")	Defines property name in user JSON profile schema	UI State
2	Attribute Value Field	Enter attribute value (e.g. "Computer Science & AI", "Mon-Thu 2-4 PM")	Defines property value stored with user account	UI State
3	Live JSON Metadata Inspector	Inspect real-time structured JSON document representing user metadata	Live reactive inspector confirming schema structure prior to database commit	UI State

24 Complete Staff REST API & Permissions Endpoint Matrix

Observed REST API catalog extracted from live network traffic for staff operations.

Method	Route	Accessed Resources	Permission Authority
GET/POST	/api/admin/audit	N/A	Public or Session
GET/POST	/api/admin/invite-codes	invite_codes, profiles	Public or Session
GET/POST	/api/admin/users	profiles, audit_log	Public or Session
GET/POST	/api/admin/users/search	profiles	Public or Session
GET/POST	/api/assessments	profiles, assessments, courses	Admin + Staff
GET/POST	/api/assessments/[id]/results	profiles, assessments, assessment_results, courses, students	Admin + Staff
GET/POST	/api/attendance	profiles, attendance, students, courses	Admin + Staff
GET/POST	/api/auth/login	profiles	Admin + Staff
GET/POST	/api/auth/logout	N/A	Public or Session
GET/POST	/api/auth/redeem-invite	profiles, invite_codes	Public or Session
GET/POST	/api/auth/signup	invite_codes, profiles	Admin + Staff
GET/POST	/api/auth/verify-invite	invite_codes	Public or Session
GET/POST	/api/brand	N/A	Public or Session
GET/POST	/api/certificates	profiles, students, certificates, verifiable_documents	Admin Only
GET/POST	/api/course-categories	N/A	Public or Session
GET/POST	/api/course-materials	profiles, course_materials	Admin + Staff
GET/POST	/api/course-trainers	profiles, course_trainers	Admin Only
GET/POST	/api/courses	courses	Public or Session
GET/POST	/api/gst	profiles, gst_settings	Admin Only
GET/POST	/api/health	N/A	Public or Session
GET/POST	/api/invoices/[invoice]/download	profiles, payments, verifiable_documents, students, gst_settings	Admin Only
GET/POST	/api/notifications	N/A	Public or Session
GET/POST	/api/payments	N/A	Public or Session
GET/POST	/api/ready	N/A	Public or Session
GET/POST	/api/reports/export	N/A	Public or Session
GET/POST	/api/reports	N/A	Public or Session
GET/POST	/api/session	N/A	Public or Session
GET/POST	/api/skills	N/A	Public or Session
GET/POST	/api/students	N/A	Public or Session
GET/POST	/api/user/settings	profiles	Public or Session
GET/POST	/api/verify/[code]	verifiable_documents, certificates, payments	Public or Session

25

Role Authority Matrix & Faculty Security Boundaries

Faculty permissions matrix strictly isolating sensitive administrative functions.

Module / Resource	Staff Faculty Authority	System Enforcement
Classroom Dashboard	Read / Action Docks	Full access to active batches & watchlist
Students & Admissions	Read / Create / Dossier View	Allowed: student profile creation & dossier inspection
Attendance Roll-Call	Read / Create / Upsert Marks	Full access to cohort roll-call sheets
Assessments & Quizzes	Read / Create / Grade Studio	Full access to test creation, QR projection & grading
Course Materials	Upload, Download & Delete	Full access to course learning resource repository
Academic Reports	Read / Export Attendance CSV	Allowed: classroom analytics and score rosters
Financial Ledger & Invoices	BLOCKED	HTTP 403 Forbidden / Auto-redirect to dashboard
Certificate Issuance	BLOCKED	HTTP 403 Forbidden / Auto-redirect to dashboard
User & Role Management	BLOCKED	HTTP 403 Forbidden / Auto-redirect to dashboard
Audit Forensics Ledger	BLOCKED	HTTP 403 Forbidden / Auto-redirect to dashboard

26

Faculty Keyboard Shortcuts & Rapid Roll-Call Navigation

Keyboard productivity controls designed for rapid classroom roll-call and test evaluation.

Keyboard Key	Context / Screen	Action Triggered
↑ / ↓	Attendance Marking Sheet	Navigate between student rows in the roll-call roster.
P	Selected Student Row	Quickly toggle status to Present .
A	Selected Student Row	Quickly toggle status to Absent .
L	Selected Student Row	Quickly toggle status to Late .
Ctrl + S	Attendance Hub	Trigger immediate bulk commit / Save Attendance action.
/	Global Topbar	Focus global learner search field.

27

Classroom Hardware, Tablet & Network Matrix

Faculty lab hardware, mobile tablet, and campus WiFi operating recommendations.

Classroom Equipment	Specification / Setup	Operational Recommendation
Faculty Laptop / PC	Core i5+ / 8GB RAM / Chrome 115+	Primary terminal for attendance marking and curriculum file management.
Classroom Projector	1080p HDMI or wireless Miracast	Project Classroom QR Modal on large auditorium display for quiz sync.
Instructor Tablet	iPad (iOS 16+) / Android (v12+)	Mobile roll-call on tablet touchscreen during physical lab inspections.
Classroom WiFi	5 GHz WPA2 Enterprise (WLAN)	Ensure separate faculty SSID to guarantee sub-100ms API response latency.

28

Institutional Terminology & Academic Glossary

Term	Standard Definition
Register ID	Permanent student registration number assigned during institutional admission.
Attendance Watchlist	Automated risk filter isolating students with cumulative attendance under 75% for counselor follow-up.
Grading Studio	Integrated evaluation worksheet allowing inline score entry, Google Sheets sync, and qualitative feedback.
Classroom QR Matrix	High-contrast 2D barcode dynamically generated from Google Form URLs for instant student quiz access.
Presigned Cloud URL	Cryptographically signed, temporary download link granting secure access to cloud course files.
Single-Use OTP Code	Disposable invite token (STAFF-XXXX-YYYY) required for initial instructor account registration.

29

Faculty Classroom Emergency Playbook & Quick-Fix Matrix

Classroom Emergency	Immediate Action	System Recovery Step
Projector Fails to Display QR	Projector HDMI cable or input mismatch.	Click "Copy Link" button in Classroom QR modal → share direct link via classroom chat channel.
Campus WiFi Outage Mid-Session	Local router disconnected or ISP disruption.	Keep attendance marking tab open. Marks remain in client memory → click Save Attendance once connection restores.
Student Absent Without Prior Notice	Unexcused learner absence in morning roll-call.	Mark student as "Absent" → check Watchlist at 05:00 PM → click phone icon to trigger counselor contact link.
Google Sheet Response Sync Error	Spreadsheet permissions set to private or URL invalid.	Verify spreadsheet is shared with institutional domain or enter scores directly via inline grading inputs.
Accidental Wrong Status Marked	Instructor accidentally marked student Absent instead of Present.	Click correct status button ("Present") → click "Save Attendance" → verify updated timestamp in Audit Log tab.